

AWS Cost Estimator

Pricing & Infrastructure Report

Bank-Name

Mode: SaaS — BusinessNext
Hosted

Date: 13 March 2026

Inflation: 4% / yr

AWS Region: us-east-1

Executive Summary

High-level cost overview and infrastructure snapshot

\$11,924	\$143,083	\$774,984	15,500
Monthly Cost	Annual Cost	5-Year Total	Named Users
Production + environments	Year 1 total	At 4% inflation/yr	Year 1 licensed

Monthly	Annual	5-Year
\$11,924	\$143,083	\$774,984

Project Details

Customer / Bank	Bank-Name
Client Mode	SaaS — BusinessNext Hosted
Database Type	PostgreSQL
Total Named Users	15,500
Concurrent Users	4,650 (30% of named)
Mobile Users	4,050
Worker Nodes	9
Total vCPUs (worker)	96
Total RAM GB (worker)	172 GB
Data Size	2100 GB
Estimate Date	13 March 2026

Node Distribution

AI-assisted infrastructure architecture

Total Worker Nodes	9
Total DB Nodes	13
Fixed / Managed Roles	7
Confidence	High
Distribution Method	Groq LLM + Rule-based

Category	Role / Service	Nodes	Instance Fam.	vCPU/node	RAM GB/node	Model
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	Compute Intensive	8	16	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	Compute Intensive	8	16	Reserved 3 Yr
PGSQL Database	Linux Nodes – Database server (Primary)	2	Memory Intensive	32	128	None
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	Memory Intensive	2	8	None
PGSQL Database	Storage: SAN (Primary DB)	2	10K IOPS	—	—	None
DMS	DMS Storage + Replication	1	Cloud Storage Service	—	—	On Demand
PGSQL Database – Reporting	Linux Nodes – Database server (Reporting)	1	Memory Intensive	32	128	None
PGSQL Database – Reporting	Storage: SAN (Reporting DB)	1	10K IOPS	—	—	None
Security & Compliance	WAF + AWS Shield Advanced	1	Managed Service	—	—	On Demand
Security & Compliance	CloudTrail + Config Audit Logs	1	Managed Service	—	—	On Demand
Kubernetes Cluster	Cloud Managed K8s	1	K8s Service	—	—	None
Caching & Session	Elastichache Service	2	Memory Optimized	4	16	Reserved 3 Yr
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	—	—	On Demand
Infrastructure & Monitoring	Bastion Host	1	General Purpose	2	4	Reserved 3 Yr
Infrastructure & Monitoring	NAT Gateway	2	NAT	—	—	On Demand
Infrastructure & Monitoring	Image Registry (ECR)	1	Managed Service	—	—	On Demand
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage Service	—	—	On Demand

AWS Cost Breakdown

Service-by-service monthly and annual costs

Compute & Database Roles

Category	Role	Nodes	Instance	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	m5.4xlarge	\$0.77	\$3,487	\$0
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	m5.2xlarge	\$0.38	\$321	\$0
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	m5.2xlarge	\$0.38	\$724	\$0
PGSQL Database	Linux Nodes – Database server (Primary)	2	r5.8xlarge	\$2.02	\$2,991	\$0
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	r5.large	\$0.13	\$400	\$0
PGSQL Database	Storage: SAN (Primary DB)	2	EBS io2	\$0.00	\$525	\$0
DMS	DMS Storage + Replication	1	EBS io2	\$0.00	\$738	\$0
PGSQL Database – Reporting	Linux Nodes – Database server (Reporting)	1	r5.8xlarge	\$2.02	\$1,496	\$0
PGSQL Database – Reporting	Storage: SAN (Reporting DB)	1	EBS io2	\$0.00	\$262	\$0
Kubernetes Cluster	Cloud Managed K8s	1	EKS	\$0.10	\$73	\$0
Caching & Session	ElastiCache Service	2	cache.r6g.large	\$0.33	\$482	\$0
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	\$0.00	\$33	\$0
Infrastructure & Monitoring	Bastion Host	1	m5.large	\$0.10	\$94	\$0
Infrastructure & Monitoring	NAT Gateway	2	NAT	\$0.00	\$41	\$0
Infrastructure & Monitoring	Image Registry (ECR)	1	ECR	\$0.00	\$51	\$0
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	S3	\$0.00	\$118	\$0
Operations	CloudWatch Monitoring	0	CloudWatch	\$0.00	\$87	\$0

Cost Summary

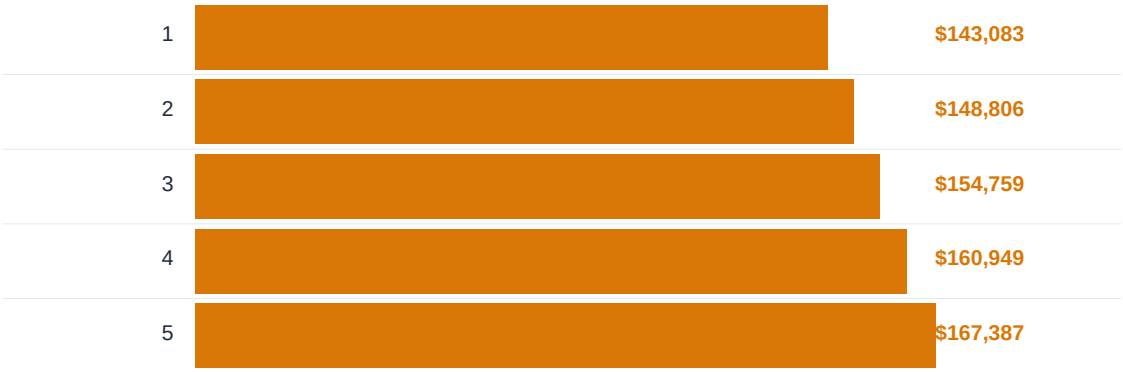
Total Monthly (Production)	\$11,924
Total Annual (Year 1)	\$143,083
5-Year Total (with inflation)	\$774,984
Inflation Rate Applied	4%

5-Year Inflation Forecast

Cost projection at 4% annual inflation

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative Total
Year 1	1.0000×	\$11,924	\$143,083	\$143,083
Year 2	1.0400×	\$12,401	\$148,806	\$291,889
Year 3	1.0816×	\$12,897	\$154,759	\$446,648
Year 4	1.1249×	\$13,412	\$160,949	\$607,597
Year 5	1.1699×	\$13,949	\$167,387	\$774,984

Annual Cost Trend



Total 5-Year Investment	\$774,984
Average Annual Cost	\$154,997
Average Monthly Cost	\$12,916
Year-over-Year Growth	4.0% (inflation) applied from Year 2

Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

Pre-Prod / SIT / UAT Environment

Monthly	Annual	5-Year
\$3,495	\$41,939	\$0

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI)	2	m5.4xlarge	\$1,169
Kubernetes	Worker Nodes (Grafana & Prometheus) – optional	1	m5.2xlarge	\$321
Kubernetes	Worker Nodes (EFK) – optional	1	m5.4xlarge	\$602
PGSQL DB	DB Server (Pre-Prod)	1	m5.4xlarge	\$585
PGSQL DB	Storage: SAN Pre-Prod	1	EBS io2	\$125
PGSQL DB	Storage: SAN SIT	1	EBS io2	\$62
PGSQL DB	Storage: SAN UAT	1	EBS io2	\$62
DMS	DMS Storage + Replication	1	dms.c5.large	\$147
Caching	Caching Service (ElastiCache)	1	cache.r6g.large	\$121
Infrastructure	Load Balancer (Internal + External)	1	ALB	\$16
Infrastructure	NAT Gateway	2	NAT	\$32
Infrastructure	Bastion Host	1	m5.large	\$94
Infrastructure	Image Registry (ECR)	1	ECR	\$50
Backup	Back Up – DB & Infra Logs	1	S3	\$34

Disaster Recovery (DR) Environment

Monthly	Annual	5-Year
\$9,504	\$114,052	\$642,454

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (DR)	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – DR	4	m5.4xlarge	\$2,324
Kubernetes	Worker Nodes (Grafana) – DR	2	m5.2xlarge	\$643
Kubernetes	Worker Nodes (EFK) – DR	2	m5.2xlarge	\$724
PGSQL DB	DB Server Primary – DR	2	m5.8xlarge	\$2,291
PGSQL DB	DB Cluster (etcd+haproxy) – DR	4	m5.large	\$312
PGSQL DB	Storage: SAN Primary – DR	2	EBS io2	\$750

Category	Role	Nodes	Instance	\$/month
PGSQL DB	Reporting DB – DR	1	m5.8xlarge	\$1,145
PGSQL DB	Storage: SAN Reporting – DR	1	EBS io2	\$375
DMS	DMS Storage + Replication – DR	1	dms.c5.large	\$296
Caching	Elastichache Service – DR	2	cache.r6g.large	\$242
Infrastructure	Load Balancer – DR	2	ALB	\$33
Infrastructure	NAT Gateway – DR	2	NAT	\$32
Infrastructure	Bastion Host – DR	1	m5.large	\$94
Infrastructure	Image Registry (ECR) – DR	1	ECR	\$51
Backup	Back Up – DR	1	S3	\$118

PUPM Analysis

Price Per User Per Month — 5-year projection

Year	Named Users	Monthly Usage	Platform Annual	AWS Total	Discounted Cost	PUPM
Year 1	15,500	\$25,931	\$323,350	\$339,517	\$402,611	\$2.1646
Year 2	16,275	\$26,928	\$335,781	\$352,570	\$407,923	\$2.0887
Year 3	17,088	\$27,965	\$348,709	\$366,144	\$423,629	\$2.0659
Year 4	17,943	\$29,043	\$362,154	\$380,262	\$439,963	\$2.0433
Year 5	18,840	\$30,165	\$376,138	\$394,945	\$456,951	\$2.0212

Discounted Annual Cost Trend

1		\$402,611
2		\$407,923
3		\$423,629
4		\$439,963
5		\$456,951

Formula Applied

Business Support	3.91% of Total Monthly Usage
Buffer	5% of Total Platform Annual Cost
Managed Services	30% of Total AWS Annual Cost
Discount	11% off Total Cost
PUPM	Discounted Annual Cost ÷ 12 ÷ Named Users
User Growth	+5% per year (Year-over-Year)
One-Time Costs (Y1)	Performance: \$5,000 · Migration: \$5,000 · Managed Setup: \$1,000

GCP Compute Engine Pricing

Google Cloud equivalent — US Central (Iowa)

Monthly	Annual	5-Year
\$18,327	\$219,923	\$1,191,176

Category	Role	Nodes	GCP Machine	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	n2-standard-16	\$1.17	\$5,378	\$64,540
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	n2-standard-8	\$0.58	\$513	\$6,162
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	n2-standard-8	\$0.58	\$1,201	\$14,412
PGSQL Database	Linux Nodes – Database server (Primary)	2	n2-highmem-32	\$3.86	\$5,734	\$68,804
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	n2-highmem-2	\$0.24	\$772	\$9,263
PGSQL Database	Storage: SAN (Primary DB)	2	Persistent Disk Extreme	\$0.00	\$525	\$6,300
DMS	DMS Storage + Replication	1	Persistent Disk Extreme	\$0.00	\$738	\$8,850
PGSQL Database – Reporting	Linux Nodes – Database server (Reporting)	1	Cloud SQL Postgres (32vCPU/128GB)	\$3.02	\$2,256	\$27,074
PGSQL Database – Reporting	Storage: SAN (Reporting DB)	1	Persistent Disk Extreme	\$0.00	\$262	\$3,150
Kubernetes Cluster	Cloud Managed K8s	1	GKE Standard Cluster	\$0.10	\$73	\$876
Caching & Session	Elastichache Service	2	Memorystore Redis (redis-16gb)	\$0.24	\$358	\$4,292
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	Cloud Load Balancing	\$0.00	\$35	\$420
Infrastructure & Monitoring	Bastion Host	1	n2-standard-2	\$0.15	\$107	\$1,279
Infrastructure & Monitoring	NAT Gateway	2	Cloud NAT	\$0.00	\$32	\$384
Infrastructure & Monitoring	Image Registry (ECR)	1	Artifact Registry	\$0.00	\$51	\$614
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage (Nearline)	\$0.00	\$205	\$2,458
Operations	Cloud Operations (Logging + Monitoring)	0	Cloud Operations Suite	\$0.00	\$87	\$1,044

GCP 5-Year Forecast

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative
Year 1	1.0000×	\$18,327	\$219,923	\$219,923
Year 2	1.0400×	\$19,060	\$228,720	\$448,644
Year 3	1.0816×	\$19,822	\$237,869	\$686,513
Year 4	1.1249×	\$20,615	\$247,384	\$933,896
Year 5	1.1699×	\$21,440	\$257,279	\$1,191,176

AWS vs GCP Cost Comparison

Side-by-side cost analysis — same workload, two clouds

\$11,924	\$18,327	\$6,403	AWS
AWS Monthly	GCP Monthly	Monthly Diff	5-Yr Winner
us-east-1	us-central1	AWS is cheaper	Saves \$416,192

AWS Annual	\$143,083
GCP Annual	\$219,923
AWS 5-Year Total	\$774,984
GCP 5-Year Total	\$1,191,176
Cheaper (Monthly)	AWS
Cheaper (5-Year)	AWS
5-Year Saving	\$416,192

Category-Level Breakdown

Category	AWS \$/mo	GCP \$/mo	Difference	% Diff	✓ Cheaper
Caching & Session	\$482	\$358	\$124	34.7%	GCP
DMS	\$738	\$738	\$0	0.0%	AWS
Infrastructure & Monitoring	\$337	\$430	\$92	21.5%	AWS
Kubernetes Cluster	\$4,605	\$7,166	\$2,560	35.7%	AWS
Operations	\$87	\$87	\$0	0.0%	AWS
PGSQL Database	\$3,916	\$7,031	\$3,114	44.3%	AWS
PGSQL Database – Reporting	\$1,758	\$2,519	\$760	30.2%	AWS
Security & Compliance	\$0	\$0	\$0	0.0%	AWS

5-Year Year-by-Year Comparison

Year	AWS/mo	GCP/mo	AWS/yr	GCP/yr	AWS Cumul.	GCP Cumul.	Winner
Year 1	\$11,924	\$18,327	\$143,083	\$219,923	\$143,083	\$219,923	AWS
Year 2	\$12,401	\$19,060	\$148,806	\$228,720	\$291,889	\$448,644	AWS
Year 3	\$12,897	\$19,822	\$154,759	\$237,869	\$446,648	\$686,513	AWS
Year 4	\$13,412	\$20,615	\$160,949	\$247,384	\$607,597	\$933,896	AWS
Year 5	\$13,949	\$21,440	\$167,387	\$257,279	\$774,984	\$1,191,176	AWS

Notes & Assumptions

AWS Pricing	All prices are AWS On-Demand, us-east-1 region, Linux/UNIX, shared tenancy. Reserved or Savings Plans pricing will reduce costs by 30–40%.
Inflation Rate	4% per year applied to all compute costs from Year 2 onwards.
Fallback Rates	If live AWS Pricing API is unavailable, hardcoded March 2026 estimates are used.
Node Distribution	Distribution computed via rule-based algorithm enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 cache prevents redundant calls.
Security Costs	Airtel SOC (\$500/mo), SOC Machines (Rs 30,000/yr), Req 4 Antivirus (Rs 2,600/yr), Req 6 Data Discovery (Rs 4,000/yr) are fixed INR-converted estimates.
Pre-Prod / DR	Pricing derived from reference sizing workbook. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server and Oracle use AWS RDS Managed.
Named Users	Year-over-year growth assumed at 5%. Concurrent users assumed at 30% of named users.
GCP Pricing	GCP prices are estimates using hardcoded us-central1 On-Demand rates (March 2026). GCP Cloud Billing API integration available via GOOGLE_CLOUD_PROJECT env var.
Confidentiality	This document contains confidential cost estimates. Do not distribute outside the project team without authorisation.

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